

and deposited are the subject of legitimate scientific debate, the acid precipitation research needed to answer all the questions raised by this debate is not proceeding at a pace which will guarantee clear answers before serious harm to life-supporting ecosystems occurs. Once the limited buffering capacities of the mountain areas have been exhausted, damage done thereafter will be irreversible.

Wise public policy suggests that the pace of measurement and research should be increased so as to permit conclusions to be drawn as quickly as possible.

But during those years necessary to obtain sufficient data to draw scientifically reliable conclusions about the rate of deposition, and the strategies needed to reduce deposition to harmless levels, policies should be adopted to minimize existing emissions of acid precursor pollutants to the extent consistent with the use of available control technology on the most significant sources of those pollutants in the West, i.e., autos, smelters, and power plants.

V.—RECOMMENDATIONS

1. The most urgent need is for better information regarding the geographical distribution and rate of acid deposition in the West. Better information means both expanding current sampling programs to include areas not now being studied, and the use of standardized sampling and analytical procedures to ensure the maximum usefulness of data now being collected.

To achieve an expanded program, EDF strongly recommends that the environmental protection programs of the three major federal land management agencies be expanded to include an integrated, interagency sampling program to better characterize total acid deposition in various regions, and the development of water quality data measurement system capable of identifying long-term changes in acidity and mineral leaching.

The Forest Service and National Park Service should be immediately funded for the express purpose of establishing and maintaining acid precipitation sampling programs in parks and wilderness areas throughout the West which are designated Class I under the Clean Air Act.

Currently, the Parks Service maintains a limited number of sampling stations in a few large parks along the Continental Divide. This program should be expanded to include other national parks and monuments which would help develop a wider distribution of samples under more diverse conditions, and with varying proximity to major sources of acid-precursor pollutants.

The Forest Service has initiated efforts to establish an acid precipitation station in Colorado in an area likely to be affected by new emissions of acid-forming pollutants from oil shale retorts.

Other wilderness areas should also be included in a broader effort by the Forest Service to measure acid deposition.

Expanded efforts to develop baseline data in Class I areas are absolutely essential if these management agencies are to have the capability to carry out their duty to protect "air quality related values" under section 165(d) of the Clean Air Act.

2. Major sources of acid-precursor pollutants—SO₂ and NO_x—should be controlled to minimize total atmospheric loadings of those pollutants until trends in the rate of acid deposition can be established.

Given the potentially high economic, environmental and social costs of widespread destruction of ecosystems and contamination of water supplies that may result from current levels of acid deposition in the West, it is prudent public policy to minimize the possible adverse effects until better data are available to more accurately assess the risks of substantial harm.

Focusing on the most significant sources of

NO_x and SO₂ emissions, the most effective strategies for minimizing these pollutants include:

(1) Retaining the existing statutory standards for the emission of NO_x from motor vehicles (section 202), rather than doubling the standards as proposed by the automobile industry. Production of vehicles complying with standards now in the law will achieve a 40 percent reduction in total NO_x emissions from mobile sources by 1990. See National Commission on Air Quality Report, Table 21, p. 198. Enactment of the auto industry proposals will result in maintaining or slightly increasing total loadings by 1990. The 1981 auto fleet meets the existing 1.0 gr/ml NO_x standard. New technology is not required.

(2) Retain the existing requirement that all new sources of pollution be required to use "best available control technology" (section 165). Fluidized bed combustion technology and other techniques for reducing NO_x emissions from power plants and other major stationary sources of NO_x should be required as best available control technology for new sources. Experimental applications of fluidized bed technology have shown that major reductions in total NO_x emissions can be achieved, whereas now stationary sources are not required to reduce NO_x emissions.

(3) SO₂ emissions from smelters in the West should be required to comply with state implementation plans. Current SO₂ emissions from smelter exceed 1.1 million tons of SO₂ per year, and account for about 80 percent of all SO₂ emissions * * * from smelters to less than 400,000 tons per year. The technology now exists to achieve the levels of control required by state plans. But these reductions can be achieved only if the compliance exemptions enacted in 1977 (section 119) are repealed.

3. The duty of Federal Land Managers and the states under section 165(d) of the Act to protect "air quality related values" in national parks and wilderness created before 1977 should be extended to new national park and wilderness areas. The "air quality related values" test for federal lands is the only provision which gives the government authority to regulate emissions which will cause harm by the deposition of acids.

4. The protections against visibility impairment in parks and wilderness areas should also be retained because EPA research shows that the sulfates and nitrates which cause acid rain are also the pollutants responsible for visibility impairment. Protecting visibility may provide substantial protection against acid deposition.

VI.—RESEARCHERS CONTACTED FOR THE SURVEY

NADP and the Fish and Wildlife Service Rocky Mountain Assessment: Dr. James Gibson, C.S.U., Fort Collins, Colorado, 491-5571.

USGS: Colorado, John Turk, 234-3467; New Mexico (water runoff study), Kim Ong, (505) 766-2011; Idaho, Tom Edwards, (208) 334-1750; and Montana, Roger Napton, (406) 440-5263.

NPS: Colorado Rocky Mountain National Park Study, Jill Baron, 491-5642; Colorado, General Information, Ray Herman, 491-5642.

National Soil Survey Laboratory, Lincoln, Nebraska: George Hohmgren, (402) 471-5363. University of Wyoming: Carol Baird, (307) 766-6732.

University of Colorado: Dr. Michael Grant, 492-7869; Dr. William Lewis, 492-6378; Dr. James Halpenny, 492-8841; and Dr. Patrick Weber (Niwot Ridge Study), 492-7909.

NCAR: Allan Lazarus, 494-5151.

University of California project at Gothic, Colorado: John Harte, Rocky Mountain Biological Lab, Crested Butte, Colorado 81224; Lawrence Berkeley Laboratory, University of California, Berkeley, California 94720, (415) 486-6119.

APPENDIX I (Memorandum)

ENVIRONMENTAL DEFENSE FUND,

August 10, 1981.

To: Concerned Officials, Organizations and Individuals.

From: William Y. Brown, Scientists, Wildlife Program, Michael J. Bean, Attorney, Wildlife Program.

Subject: Acid Rain and Wildlife Conservation.

Enclosed is an EDF paper entitled "Acid Rain and Wildlife Conservation." The paper distills a mass of information compiled jointly by the United States and Canada in preparation for the negotiation of an international agreement on transboundary air pollution. Although prepared for internal governmental use, these data were recently made available to the public. Several additional studies, including in particular work done on damage to forests, are also included in the EDF paper.

The observations summarized in this paper are sobering. Major areas of eastern North America, and perhaps the West, are threatened with massive loss of life forms because of acid rain. Although continuing, intensive research on the problem is essential, failure to take immediate steps to curtail the responsible pollutants, sulphur dioxide in particular, would demonstrate a reckless attitude of our society toward living resources.

We urge you to read the paper carefully and to do what you can to solve the problem. Two fundamental steps are urgent:

The Congress must not weaken the limits on sulphur and nitrogen oxide pollution currently required under the Clean Air Act.

A new law should be enacted to substantially reduce sulphur dioxide emissions in the eastern United States. The National Clean Air Coalition is working on this bill and it needs your support.

Please contact us if we can be of further assistance.

ACID RAIN AND WILDLIFE CONSERVATION

(By Dr. William Y. Brown)

Acid deposition is currently being observed in most of eastern North America. Within this half-continent are large areas in which the surface soil material and bedrock types have little buffering capacity for acid inputs and are identified as "potentially sensitive." These areas include some of the most unique unspoiled and biologically productive environments in North America. The potential is high for environmental degradation from the deposition of acid and other pollutants. (U.S.-Canada Work Group on Impact Assessment of Transboundary Air Pollution, Interim Report, January 15, 1981.)

THE CAUSES

Acid rain is caused principally by pollution of the air with sulphur oxides and nitrogen oxides. Most sulphur oxides come from the smoke of coal-fired power plants and ore smelters. These power plants, plus car and truck exhaust, add nitrogen oxides to the air. In atmospheric water, sulphuric acid and nitric acid are formed. These acids fall with rain and snow, then flow to streams and lakes. Precipitation in large areas of North America is forty times more acid than normal. Sometimes combined with acids formed on the ground by gaseous and solid pollution, this acid rain is destroying wildlife in poorly buffered areas over much of the eastern United States and Canada. (1,2)

THE MECHANISMS

Acid rain destroys wildlife primarily in three ways. First, acid rain directly inhibits reproduction in many species. (2,3,4) Second, acid rain can release toxic metals (aluminum, manganese, iron) which kill animal and plant life. (5,6,7,8,9) Third, acid rain can discharge from ecosystems other metals which

are essential to many plants and animals (calcium and magnesium). (8,9)

THE DESTRUCTION OF WILDLIFE

Freshwater lakes and streams

Acid rain can dramatically alter the kinds and diversity of animals and plants inhabiting lakes and streams. Many lakes have been stripped of most native life altogether.

Disruption of microorganisms

Field and laboratory studies in Scandinavia and North America indicate that bacterial diversity decreases in acidified water, accompanied by an increase in fungi, decrease in community respiration, and possible inhibition of the Nitrogen Cycle. (2,10)

Degradation of plankton

Plankton communities in lakes are sensitive to rising acid concentration. Generally one or two species of resistant algae become abundant while other native species diminish and may be lost. (11, 12) Overall, species diversity drops. (13, 14, 15) The dominant algae in acid lakes, a group called diatoms, cannot readily be eaten by invertebrate animals in the plankton. (2, 11)

Invasion of Sphagnum moss

When lakes become acid, species of larger plants may be replaced by Sphagnum mosses. Moss invasion has been documented in Sweden, Canada, and New York State. (16, 17, 18, 19) Sphagnum moss forms extensive, dense beds. In New York State, growth of moss has been accompanied by proliferation of acid tolerant algae, which has completely covered larger plants in some areas and blocked the capture of sunlight. (19)

Disappearance of molluscs and crustaceans

Molluscs and crustaceans cannot live in acid waters. Molluscs are generally restricted to water with pH greater than 5.8 to 6.0—less acidic than many lakes affected by acid rain. (20, 21, 22) Many crustaceans are similarly sensitive. (23) The need of these animals for calcium, and its depletion from waters turned acid, may be responsible. (2)

Failure of reproduction in amphibians

Many species of frogs, toads and salamanders breed in temporary pools of meltwater formed in the spring. These pools are often very acid. Embryonic mortality and deformity is great in certain salamanders at pH less than 6.0. Yet the average breeding pool for these same species is 32 times more acid. (3, 24) Particularly sensitive species of amphibians include: (25)

- Wood frog (*Rana sylvatica*).
- American toad (*Bufo americanus*).
- Northern spring peeper (*Hyla crucifer*).
- Chorus frog (*Pseudacris triseriata*).
- Yellow-spotted salamander (*Ambystoma maculatum*).
- Blue-spotted salamander (*Ambystoma laterale*).
- Tremblays salamander (*Ambystoma tremblayi*).

It is inevitable that amphibian species such as these will suffer severe decreases and will become locally extinct. Reduction in the biomass of amphibians may have far-reaching effects in both aquatic and terrestrial ecosystems, because salamanders and frogs are important prey for birds and mammals. (2)

Fish loss

High acid concentration can inhibit larval development of fish and promote aluminum and mercury concentrations to toxic levels. (2, 5, 7, 26-30) Massive kills have been observed for Atlantic salmon (31) and brown trout (32) in Europe and for pumpkinseeds, during spring snow melt, in North America. (33) However, less dramatic increases in mortality and drops in reproduction are more significant over the long term, eliminating

many species from significant areas of their range:

Brook trout no longer can survive in at least 180 acidified Adirondack ponds where they formerly lived. (34)

Nearly half of the Adirondack lakes may be acidified and devoid of fish—10 times more than 50 years ago. (35)

Fish have disappeared from many lakes in Ontario, Canada. Consider Lumsden Lake (one of 68 case studies): (33)

1950s: 8 species of fish present.
1960: Last report of yellow perch and burbot.

1960-65: Sport fishery falls (pH 6.8 in September of 1961).

1967: Last capture of lake trout.
1968: White sucker suddenly rare.

1969: Last capture of trout perch, lake herring and white sucker.

1970: Only lake chub present.

1971: Lake Chub very rare (pH 4.4 in August of 1971).

Thousands of lakes in Ontario and Quebec, many of them prime sport fisheries for U.S. and Canadian citizens, may be similarly lost if acid rain continues at present rates. (2)

In Nova Scotia, acidification threatens 30% of the Atlantic salmon populations. Salmon reproduction has ceased in nine rivers where the fish once lived. Canada may be following in the footsteps of Norway, where acid rain has devastated salmonid fish. (36)

Fish populations in lakes and rivers generally are genetically distinct, being locally adapted. Elimination of a population is therefore not reversible by restocking. Furthermore, replacement stocks will not survive if acid rain is not abated, nor will they survive with lower acid concentrations if essential elements have been expelled from the ecosystem by former high acidity. (2)

Decline of birds and mammals

Birds and mammals which depend on acid-sensitive animals and plants for food can be expected to decline when feeding lakes and streams become acidified. Fish-eating birds have disappeared from acid lakes in Sweden. (14, 37) In North America, the Common Loon has declined in the Adirondacks over the last 15 years, paralleling the decline in fish populations as well as increased human disturbance. (38, 39) The following birds and mammals are considered particularly susceptible to acid rain in North America and should be monitored closely:

Birds and Mammals Susceptible to Acid Rain (25)

- Common Loon (*Gavia immer*).
- Common merganser (*Mergus merganser*).
- Hooded merganser (*Mergus cucullatus*).
- Red-breasted merganser (*Mergus serrator*).
- Great blue heron (*Ardea herodias*).
- Belted kingfisher (*Megasceryle alcyon*).
- Common goldeneye (*Brachypteryx alpestris*).
- Ring-necked duck (*Aythya collaris*).
- Black Duck (*Anas rubripes*).
- Green-winged teal (*Anas carolinensis*).
- Mallard (*Anas platyrhynchos*).
- Northern pintail (*Anas acuta*).
- American widgeon (*Anas americana*).
- American mink (*Mustela vison*).
- River otter (*Lutra canadensis*).
- Muskrat (*Ondatra zibethicus*).

Forests

Beech and spruce forests in West Germany provide the only long-term information concerning effects of acid rain on forests. (8) In these forests, already acidic rain is made two to four times more acid by sulphur dioxide and particulates. The acid is releasing aluminum salts in the soil, killing rootlets. The beech forest is considered to be doomed and the spruce forest unlikely to survive.

These studied West German forests are grown on sensitive, unbuffered soil. No industry is in the forest area, although the Essen-Düsseldorf Industrial Area lies 130-

150 km to the west. No study has demonstrated such damage to forests in North America.

However, many forest ecosystems should be considered at risk and monitored carefully. These include such unique ecosystems at the Boundary Waters Canoe Area Wilderness of Northern Minnesota, (40) the Great Smoky Mountains National Park in Tennessee and North Carolina, (41) the Adirondack Mountains in New York, (42) and the Kejimikujik National Park in Nova Scotia. (2)

THE SOLUTION TO THE PROBLEM

Acid rain will abate only with reduction in the amount of sulphur and nitrogen oxides put into the air by coal-fired power plants, smelters, refineries, cars, trucks, and other sources.

A major positive and feasible step would be reduction in the amount of sulphur dioxide emissions by power plants. The National Commission on Air Quality has estimated that a seven million ton reduction in annual sulphur dioxide emissions by eastern utilities can be achieved by 1990 with an average increase of less than two percent in eastern utility bills.

The National Clean Air Coalition is investigating the additional cost of a reduction totaling ten millions tons. Because a reduction of seven to ten million tons of sulphur dioxide constitutes some 40 to 57 percent of total emissions in the East, the abatement of acid rain would be substantial.

To stop the destruction of wildlife by acid rain, concerned citizens, organizations, government agencies and elected officials should:

Support those existing limitations on emissions of sulphur and nitrogen oxides which have been set under the Clean Air Act. Many efforts are now underway to weaken the Clean Air Act this autumn, and your help is needed to keep the law intact.

Support the efforts of the National Clean Air Coalition to have new legal authority enacted to substantially reduce sulphur dioxide emissions by 1990.

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● Mr. STAFFORD. Mr. President, I am pleased to cosponsor Senator MITCHELL's bill.

Three years ago, I was somewhat doubtful that there was a phenomenon called acid rain.

Two years ago, I believed in acid rain, but was not certain as to its causes.

One year ago, I was persuaded not only that it existed, but that acid rain was caused by sulfur and nitrogen oxides emitted primarily from utility and industrial boilers, especially in the Mid-

west. What I doubted was the will of Congress to deal with the issue.

Today, I believe the Congress has acquired not only the knowledge, but the will. Whatever happens to the Clean Air Act, it is becoming increasingly certain that the Congress will have no choice but to deal with the acid rain issue.

I know that Senator MITCHELL has developed this proposal over the course of several months and compliment him on his efforts. He has made every attempt to respond to the complaints and comments of all sides interested in this issue. He has made every attempt to develop a bill which all sides can support, for which he deserves great credit. Rather than responding to a sensitive political and environmental issue in an unthinking and rigid fashion, he has attempted to develop a balanced and reasonable bill.

Mr. President, I hope that during this session the Committee on Environment and Public Works, the Senate, and the Congress can address the acid rain problem. I believe Senator MITCHELL's bill is an excellent foundation for our work, and I am pleased to join him in its introduction. ●

Mr. CHAFEE. Mr. President, I suggest the absence of a quorum.

The PRESIDING OFFICER. The clerk will call the roll.

The legislative clerk proceeded to call the roll.

Mr. MITCHELL. Mr. President, I ask unanimous consent that the order for the quorum call be rescinded.

The PRESIDING OFFICER. Without objection, it is so ordered.

Mr. MITCHELL. Mr. President, I yield back the remainder of my time.

CONCLUSION OF MORNING BUSINESS

The PRESIDING OFFICER. All time having been yielded back, morning business is closed.

RECESS UNTIL 2 P.M.

Under the previous order, the Senate will stand in recess until 2 p.m.

Thereupon, at 1:05 p.m., the Senate recessed until 2 p.m.; whereupon, the Senate reassembled when called to order by the Presiding Officer (Mr. KASTEN).

TELECOMMUNICATIONS COMPETITION AND DEREGULATION ACT OF 1981

The PRESIDING OFFICER. The clerk will state the pending business before the Senate.

The legislative clerk read as follows:

A bill (S. 898) to amend the Communications Act of 1934 to provide for improved domestic telecommunications, and for other purposes.

Mr. PACKWOOD. Mr. President, I suggest the absence of a quorum.

The PRESIDING OFFICER. The clerk will call the roll.

The bill clerk proceeded to call the roll.

Mr. BRADLEY. Mr. President, I ask unanimous consent that the order for the quorum call be rescinded.

The PRESIDING OFFICER. Without objection, it is so ordered.

Mr. BRADLEY. Mr. President, in a few minutes I will send an amendment to the desk and ask for its immediate consideration. This is an amendment which deals with the exchange area section of the bill. It is one, I think, that is consistent with the intent of the bill.

If an exchange area is altered because of growth, the bill states that it shall be reduced to the SMSA, which, in certain States, has the effect of converting what are presently local calls into long-distance calls, dramatically increasing costs to the ratepayer. This particularly is true in densely populated States like New Jersey.

The amendment I am offering would give an option to use the standard consolidated statistical areas as well as the SMSA's.

I have cleared this amendment with the chairman.

Mr. PACKWOOD. Mr. President, indeed the Senator from New Jersey has a very unique geographical situation, and this amendment cures what would otherwise be a hardship in the telephone exchange system in New Jersey. I believe both the majority and minority have seen the amendment and we are prepared to accept it.

UP AMENDMENT NO. 466

(Purpose: To include "consolidated statistical areas" for densely populated areas)

Mr. BRADLEY. Mr. President, I send the amendment to the desk and ask for its immediate consideration.

The PRESIDING OFFICER. (Mr. COHEN). The Chair is advised that there is currently pending an amendment before the Senate by the Senator from South Carolina.

Mr. PACKWOOD. Mr. President, I just talked with the Senator from South Carolina and asked if he had any objection to temporarily laying his amendment aside.

I might say to Senator THURMOND, this is not his amendment, but it is Senator HOLLINGS' amendment.

I just talked to Senator HOLLINGS and he said he had no objection to temporarily laying his amendment aside. I ask unanimous consent to temporarily lay aside the amendment of the Senator from South Carolina so we might consider this amendment, and also ask unanimous consent that we might temporarily lay it aside to consider the next amendment, of the Senator from New Jersey.

The PRESIDING OFFICER. Without objection, it is so ordered.

The amendment will be stated.

The legislative clerk read as follows:

The Senator from New Jersey (Mr. BRADLEY) proposes an unprinted amendment numbered 466.

Mr. BRADLEY. Mr. President, I ask unanimous consent that further reading of the amendment be dispensed with.

The PRESIDING OFFICER. Without objection, it is so ordered.

The amendment is as follows:

On page 141, line 23, insert "(or a consolidated statistical area, in the case of densely populated States)" after "area".

On page 142, line 1, insert "(or a consolidated statistical area, in the case of densely populated States)" after "area".